

Tema 1

1	2	3	4	5
17	17	17	20	20-91

CALIF.

9/1000

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Introducción al Análisis Matemático
Recuperatorio Primer Parcial (29/06/16)

1. Sea $f(x) = \frac{|3x-1|+3}{\log_{1/2}(2x^2-3x+1)}$

- B a) Hallar el dominio de f . (10 pts)
R b) Hallar el conjunto de positividad de f . (10 pts)

2. Resuelva las siguientes inecuaciones

- B a) $|2x-1| \leq |x+1|$ (10 pts)
R b) $|3^{x+1} - 3^{x-2}| > 26$ (10 pts)

3. Calcule los siguiente límites

- R a) $\lim_{n \rightarrow +\infty} \frac{(3n)^n}{n!}$ (10 pts)
B b) $\lim_{n \rightarrow +\infty} \frac{2^{3n}}{9^n + 1} (-1)^n + \frac{1}{n^2 - \sqrt{4n^4 + \cos(n)}}$ (10 pts)

4. Sea k es un número real no nulo y $(a_n)_{n \in \mathbb{N}}$ la sucesión

$$a_n = \left(\frac{n^2 + \sin(2n)}{n^2} \right)^{\frac{kn^2 + 1}{n}}$$

Analizar si existe $k \neq 0$ tal que

$$\lim_{n \rightarrow +\infty} a_n \neq 1$$

(20 pts)

- B 5. Sea k un número real positivo. Mostrar tres valores distintos de k que den lugar a que el siguiente límite de tres resultados distintos.

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^3 + 2x^k + 1}}{4x^2 + 4} + e^{-kx^2 + x} \quad (20 \text{ pts})$$

JUSTIFICAR TODAS LAS RESPUESTAS

$$1) f(x) = \frac{|3x-1|+3}{\log_{1/2}(2x^2-3x+1)}$$

$$a) \text{Dom } f \begin{cases} \log_{1/2}(2x^2-3x+1) \neq 0 \\ 2x^2-3x+1 > 0 \end{cases}$$

$$\log_{1/2}(2x^2-3x+1) = 0 ?$$

$$\frac{1}{2} \log_{1/2}(2x^2-3x+1) = \frac{1}{2} \cdot 0$$

$$2x^2-3x+1 = 1$$

$$2x^2-3x = 0$$

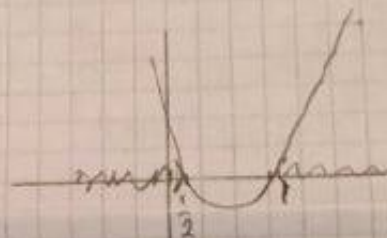
$$x=0$$

$$x=\frac{3}{2}$$

$$x \neq 0 \vee x \neq \frac{3}{2}$$

$$2x^2-3x+1 > 0$$

$$x=1 \quad x=\frac{1}{2}$$

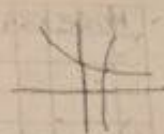


$$(-\infty, \frac{1}{2}) \cup (1, +\infty)$$

$$\text{Dom } f = \mathbb{R} \setminus \{0, \frac{1}{2}, \frac{3}{2}\}$$

$$\text{Dom } f = (-\infty, 0) \cup (0, \frac{1}{2}) \cup (1, \frac{3}{2}) \cup (\frac{3}{2}, +\infty)$$

$$B) \frac{|3x-1|+3}{\log_{1/2}(2x^2-3x+1)} > 0 \quad \begin{matrix} + & 0 & - \\ + & & - \end{matrix}$$



$$|3x-1|+3 > 0 \quad \begin{matrix} + \\ + \end{matrix}$$

$$|3x-1| > -3$$

$$3x-1 > -3$$

$$3x > -2$$

$$x > -\frac{2}{3}$$

$$|3x-1| \begin{cases} 3x-1 & \text{if } x \geq 0 \\ -3x+1 & \text{if } x < 0 \end{cases}$$

$$\begin{cases} -3x+1 < 0 \\ 3x < -1 \\ x > \frac{1}{3} \end{cases}$$

$$\log_{1/2}(2x^2-3x+1) > 0$$

⇓

$$0 < 2x^2-3x+1 < 1$$

$$\begin{cases} x < 0 \\ -3x+1 > 3 \\ x > -4 \\ x < -\frac{1}{3} \end{cases}$$

$$2x^2-3x+1 > 0$$

$$(-\infty, \frac{1}{2}) \cup (1, +\infty)$$

Ejercicio anterior

$$2x^2-3x+1 < 1$$

$$2x^2-3x < 0$$

$$x = \frac{3}{2}$$

$$x = 0$$



$$(0, \frac{3}{2})$$

$$\frac{1}{2} < x < \frac{3}{2}$$

$$(0, \frac{1}{2}) \cup (1, \frac{3}{2}) \in \text{Dom } f$$

$$|3x-1|+3 < 0$$

$$|3x-1| < -3$$

ABS

$$2) A) |2x-1| \leq |x+1|$$

$$|2x-1| \begin{cases} 2x-1 & \text{si } x \geq 0 \\ -2x+1 & \text{si } x < 0 \end{cases}$$

Caso ①

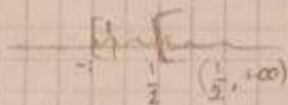
$$2x-1 \geq 0$$

$$2x \geq 1$$

$$x \geq \frac{1}{2}$$

$$x+1 \geq 0$$

$$x \geq -1$$

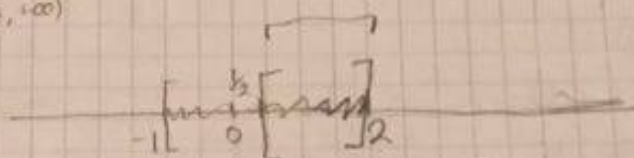


$$2x-1 \leq x+1$$

$$2x \leq x+2$$

$$x \leq 2$$

$$|x+1| \begin{cases} x+1 & \text{si } x \geq 0 \\ -x-1 & \text{si } x < 0 \end{cases}$$



$$\text{Sol ①: } \left[\frac{1}{2}, 2 \right]$$

Caso ②

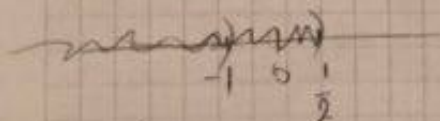
$$2x-1 < 0$$

$$2x < 1$$

$$x < \frac{1}{2}$$

$$x+1 < 0$$

$$x < -1$$



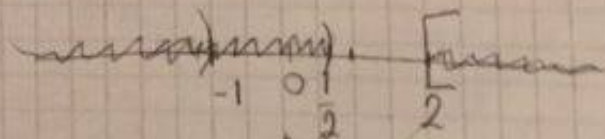
$$(-\infty, -1)$$

$$-2x+1 \leq -x-1$$

$$-2x \leq -x-2$$

$$-x \leq -2$$

$$x \geq 2$$



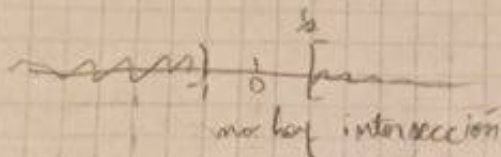
$$\text{Sol ②: } \emptyset$$

Caso ③

$$2x - 1 \geq 0 \quad \vee \quad x + 1 < 0$$

$$x \geq \frac{1}{2}$$

$$x < -1$$



$$\text{Sol} ③ = \emptyset$$

$$2x - 1 \leq -x - 1$$

$$2x \leq -x$$

$$3x \leq 0$$

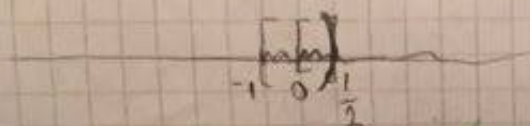
$$x \leq 0$$

Caso ④

$$2x - 1 < 0 \quad \vee \quad x + 1 \geq 0$$

$$x < \frac{1}{2}$$

$$x \geq -1$$



$$\text{Sol} ④ = \left[-1, \frac{1}{2}\right)$$

$$-2x + 1 \leq x + 1$$

$$-2x \leq x$$

$$-3x \leq 0$$

$$x \geq 0$$

$$\text{Sol} = \text{Sol} ① \cup \text{Sol} ④ = \left[\frac{1}{2}, 2\right] \cup \left[0, \frac{1}{2}\right] = [0, 2]$$

$$b) |3^{x+1} - 3^{x-2}| > 26$$

$$3^{x+1} - 3^{x-2} > 26$$

$$3^x \cdot 3 - 3^x \cdot \left(\frac{1}{3}\right)^2 > 26$$

$$3z - z - \frac{1}{9} > 26$$

$$z \frac{26}{9} > 26$$

$$z > 26 \cdot \frac{9}{26}$$

$$z > 9$$

⇓

$$3^x > 9$$

$$\log_{03} (3)^x > \log_3 (9)$$

$$x > 2$$

base > 1 no cambia la desigualdad

a

$$3^{x+1} - 3^{x-2} < -26$$

$$z \frac{26}{9} < -26$$

$$z < -9$$

$$3^x < -9$$

⇓
ABS ✓

$$3) a) \lim_{n \rightarrow +\infty} \frac{(3n)^n}{n!} \cdot \lim_{n \rightarrow +\infty} \frac{3^n \cdot n^n}{n!} \stackrel{\text{D'Alembert}}{=} \lim_{n \rightarrow +\infty} \frac{3^{n+1} \cdot n^{n+1}}{(n+1) \cdot n!} \cdot \frac{n!}{3^n \cdot n^n}$$

$$= \lim_{n \rightarrow +\infty} \frac{3n}{n+1} = \lim_{n \rightarrow +\infty} \frac{3n}{n(1+\frac{1}{n})} = 3 = l$$

$l = 3 > 1$ per lo tanto $\rightarrow +\infty$

$$\lim_{n \rightarrow +\infty} \frac{(3n)^n}{n!} = \boxed{+\infty}$$

$$b) \lim_{n \rightarrow +\infty} \frac{2^{3n}}{9^n + 1} (-1)^n + \frac{1}{n^2 - \sqrt{4n^4 + \cos(n)}} =$$

$$\lim_{n \rightarrow +\infty} \frac{8^n}{9^n + 1} (-1)^n + \frac{1}{n^2 - \sqrt{4n^4 + \cos(n)}} =$$

$$\lim_{n \rightarrow +\infty} \frac{8^n}{9^n + 1} \cdot (-1)^n + \frac{1}{n^2 - n^2 \cdot \sqrt{4 + \frac{\cos(n)}{n^4}}} =$$

$$\lim_{n \rightarrow +\infty} \underbrace{\frac{8^n}{9^n + 1} \cdot (-1)^n}_{\substack{\text{CA} \rightarrow 0 \\ \rightarrow 0}} + \frac{1}{n^2 \cdot \left(1 - \left(1 - \sqrt{4 + \frac{\cos(n)}{n^4}} \right) \cdot \frac{1}{n^4} \right)} = \boxed{0}$$

CA

$$\lim_{n \rightarrow +\infty} \frac{8^n}{9^n + 1}$$

per Cauchy
 $\rightarrow 0$

$$\lim_{n \rightarrow +\infty}$$

$$\frac{\sqrt[3]{8^{n+1}}}{\sqrt[3]{9^{n+1} + 1}} = \frac{8}{9} = l$$

$l < 1 \rightarrow 0$

$$4) \lim_{n \rightarrow +\infty} \left(\frac{n^2 + \sin(2n)}{n^2} \right)^{\frac{kn^2+1}{n}} = \lim_{n \rightarrow +\infty} \left(1 + \frac{n^2 + \sin(2n)}{n^2} - 1 \right)^{\frac{kn^2+1}{n}}$$

$$= \lim_{n \rightarrow +\infty} \left(1 + \frac{n^2 + \sin(2n) + m^2}{n^2} \right)^{\frac{kn^2+1}{n}} = \text{C.A.} \rightarrow 0$$

$$\lim_{n \rightarrow +\infty} \left[\left(1 + \frac{1}{\frac{n^2}{\sin(2n)}} \right)^{\frac{n^2}{\sin(2n)}} \right]^{\frac{\sin(2n)}{n^2} \cdot \frac{kn^2+1}{n}} = e^0 = 1$$

C.A

$$\lim_{n \rightarrow +\infty} \left(\frac{\sin(2n)}{n^2} \right) \cdot \left(\frac{kn^2+1}{n} \right) = \lim_{n \rightarrow +\infty} \left(\frac{\sin(2n)}{n^2} \right) \cdot \left(\frac{kn^2+1}{n} \right)$$

$$\frac{kn^2 \cdot \sin(2n) + \sin(2n)}{n^3} = \frac{kn^2 \cdot \sin(2n)}{n^3} + \frac{\sin(2n) \cdot \frac{1}{n^3}}{\frac{1}{n^3}}$$

$$\lim_{n \rightarrow +\infty} \frac{k \cdot \sin(2n)}{n} = k \cdot \sin(2n) \cdot \left(\frac{1}{n} \right) = k \cdot 0 = 0$$

Sea $k > 0$ o $k < 0$

siempre el resultado
no a ser 0

no existe $k \neq 0$ tal que $\lim_{n \rightarrow +\infty} a_n \neq 1$

5) $k > 0$

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^3} + 2x^k + 1}{4x^2 + 4} + e^{-kx^2+x} \rightarrow 0$$

C.A

$\lim_{x \rightarrow +\infty} e^{-kx^2+x} = e^{-\infty} = 0 \Rightarrow$ siempre para cualquier valor de $k > 0$

$\lim_{x \rightarrow +\infty} -kx^2+x = \lim_{x \rightarrow +\infty} -x^2+x = \lim_{x \rightarrow +\infty} x \cdot \left(-1 + \frac{1}{x}\right) = -\infty$

Por lo tanto me concentro en

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^3} + 2x^k + 1}{4x^2 + 4}$$

si $k=2 \Rightarrow \lim_{x \rightarrow +\infty} \frac{x^{3/2} + 2x^2 + 1}{4x^2 + 4} = \lim_{x \rightarrow +\infty} x^{1/2} \cdot \left(\frac{1}{x^{1/2}} + 2 + \frac{1}{x^2}\right) = \frac{1}{2}$

si $k=1 \Rightarrow \lim_{x \rightarrow +\infty} \frac{x^{3/2} + 2x + 1}{4x^2 + 4} = \lim_{x \rightarrow +\infty} \frac{x^{1/2} \cdot \left(1 + \frac{2}{x^{1/2}} + \frac{1}{x^{3/2}}\right)}{x^2 \left(4 + \frac{4}{x^2}\right)} = \frac{1}{+\infty} = 0$

si $k=3 \Rightarrow \lim_{x \rightarrow +\infty} \frac{x^{3/2} + 2x^3 + 1}{4x^2 + 4} = \lim_{x \rightarrow +\infty} x^{3/2} \cdot \left(\frac{1}{x^{3/2}} + 2 + \frac{1}{x^3}\right) = +\infty \cdot \frac{2}{4} = +\infty$

si $k=2 \rightarrow \frac{1}{2}$

si $k=1 \rightarrow 0$

si $k=3 \rightarrow +\infty$